

Day 16 – Shell Scripting s

Topics Covered

1. Introduction to Shell Scripting

- What is Shell Scripting?
- Advantages of Shell Scripts
- Real-world DevOps use cases
- Types of Shells (Bash, Sh, Ksh)

2. Creating Your First Script

```
#!/bin/bash  
echo "Hello World"
```

Execution Steps

```
chmod +x script.sh  
./script.sh
```

3. Variables in Shell Script

```
#!/bin/bash  
NAME="Sai"  
echo "Welcome $NAME"
```

User Input

```
#!/bin/bash  
read -p "Enter your name: " NAME  
echo "Hello $NAME"
```

4. Conditional Statements

If Statement

```
#!/bin/bash  
NUM=10
```

```
if [ $NUM -gt 5 ]
then
    echo "Number is greater than 5"
fi
```

If-Else

```
#!/bin/bash
```

```
read -p "Enter Age: " AGE
```

```
if [ $AGE -ge 18 ]
then
    echo "Eligible to Vote"
else
    echo "Not Eligible"
fi
```

5. Loops

For Loop

```
#!/bin/bash
```

```
for i in {1..5}
do
    echo $i
done
```

While Loop

```
#!/bin/bash
```

```
COUNT=1
```

```
while [ $COUNT -le 5 ]
do
    echo $COUNT
```

```
COUNT=$((COUNT+1))
done
```

6. Functions

```
#!/bin/bash

greet() {
    echo "Welcome to Shell Scripting"
}

greet
```

7. Command Line Arguments

```
#!/bin/bash

echo "First Argument: $1"
echo "Second Argument: $2"

Execution:

./script.sh AWS DevOps

Output:

First Argument: AWS
Second Argument: DevOps
```

8. File Handling

Check File Exists:

```
#!/bin/bash

FILE="/tmp/test.txt"

if [ -f $FILE ]
then
```

```
    echo "File Exists"
else
    echo "File Not Found"
fi
```

9. Useful DevOps Scripts

Disk Usage Monitoring

```
#!/bin/bash
```

```
df -h
```

Memory Usage

```
#!/bin/bash
```

```
free -m
```

Service Status Check

```
#!/bin/bash
```

```
systemctl status nginx
```

10. Backup Script

```
#!/bin/bash
```

```
SOURCE="/var/log"
```

```
DEST="/backup"
```

```
tar -czvf backup.tar.gz $SOURCE
```

```
mv backup.tar.gz $DEST
```

Hands-On Tasks

Task 1

Create a script to:

- Accept username input
- Display welcome message

Task 2

Create a script to:

- Check whether a file exists
- Print appropriate message

Task 3

Create a script to:

- Display CPU, Memory, and Disk Usage

Task 4

Create a script to:

- Take a backup of /etc
- Store backup in /backup

Interview Questions

1. What is Shell Scripting?
2. Difference between \$@ and \$*?
3. What is the purpose of #!/bin/bash?
4. Difference between for loop and while loop?
5. How do you pass arguments to a shell script?
6. How do you check if a file exists in Linux?
7. What are shell functions?
8. What is the difference between = and == in shell scripting?
9. How do you schedule shell scripts automatically?
10. How is Shell Scripting used in DevOps?